

Operating Systems — Sample Notes / Syllabus

Intentionally varied formatting to test PDF parsing robustness.

UNIT 1: Basics

- Processes; threads; process states
- CPU scheduling — FCFS, SJF, Round Robin, Priority Scheduling
- Context switching and scheduling criteria

Unit 2 Memory

- Paging and segmentation
- Virtual memory
- Page replacement: FIFO, LRU, Optimal
- Thrashing

UNIT 3 — File Systems

- File allocation methods
- Directory structures
- Disk scheduling algorithms

Important / Exam Focus

- Deadlocks: conditions, prevention, avoidance, Banker's algorithm
- Synchronization: semaphores, mutexes, critical section
- Practice numerical problems for scheduling and page replacement